The documentation and process conversion measures necessary to comply with this revision shall be completed by 10 November 2007.

INCH-POUND

MIL-PRF-19500/620G
10 August 2007
SUPERSEDING
MIL-PRF-19500/620F
18 August 2005

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, HERMETIC, DIODE, SILICON, RECTIFIER,
SCHOTTKY BARRIER, TYPES 1N5822 AND 1N5822US, 1N6864 AND 1N6864US,
JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for silicon, Schottky barrier rectifier diodes. Four levels of product assurance are provided for each device type as specified in MIL-PRF-19500, and two levels of product assurance for die (element evaluation).

1.2 Physical dimensions. See figures 1 (axial), 2 ('US' surface mount), and 3 (die) dimensions.

| Types | $V_{RWM}$ (1) (2) | $I_O$ (1) (2) | $I_{FSM}$ | $T_{STG}$ | $T_J$ (1) | $Z_{\Theta JX}$ | $R_{\Theta JL}$ .375 inch (9.52 mm) Lead length | $R_{\Theta JEC}$ |
|---|---|---|---|---|---|---|---|---|
| | V(pk) | A dc | A(pk) | °C | °C | °C/W | °C/W | °C/W |
| 1N5822, 1N5822US | 40 | 3.0 | 80 | -65 to | -65 to | 2.5 | 30 | 10 |
| 1N6864, 1N6864US | 80 | 3.0 | 80 | +150 | +125 | 2.5 | 30 | 10 |

(1) See figures 4, 5, 6 and 7 for derating curves and for effects of $V_R$ on $T_J$. The maximum $T_J$ depends on the voltage applied. $T_A$ = +75°C for both axial and Metal Electrode Leadless Face diodes (MELF) (US) on printed circuit board (PCB), PCB = FR4 - .0625 inch (1.59 mm) 1-layer 1-Oz Cu, horizontal, in still air, pads for (US) = .061 inch (1.55 mm) x.105 inch (2.67 mm); pads for axial = .092 inch (2.34 mm) diameter, strip = .030 inch (0.76 mm) x 1 inch (25.4 mm) long, lead length L ≤ .187 inch (≤4.75 mm); $R_{\Theta JA}$ with a defined PCB thermal resistance condition included, is measured at $I_O$ = 1A.

(2) $T_A$ = 55°C for both axial and MELF (US) on printed circuit board (PCB), PCB = FR4 .0625 inch (1.59mm) pad; area of each pad = .4 square inch (10.16 square mm).

Comments, suggestions, or questions on this document should be addressed to Defense Supply Center, Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to semiconductor@dscc.dla.mil . Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil .

AMSC N/A

FSC 5961

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1.4 Primary electrical characteristics. Unless otherwise specified, $T_A$ = +25 °C.

| Types | $V_{FM1}$ | $V_{FM2}$ | $V_{FM3}$ | $I_{RM}$ $V_{RM}$ = 40 V dc (1N5822) $V_{RM}$ = 80 V dc (1N6864) pulsed method (see 4.5.1) | | $R_{\Theta JL}$ .375 inch (9.52 mm) Lead length | $R_{\Theta JEC}$ |
|---|---|---|---|---|---|---|---|
| | $I_{FM}$ = 1.0 A | $I_{FM}$ = 3.0 A | $I_{FM}$ = 9.4 A | $T_J$ = +25 °C $I_{RM1}$ | $T_J$ = +100 °C $I_{RM2}$ | | |
| | V (pk) | V (pk) | V (pk) | mA | mA | °C/W | °C/W |
| 1N5822 | 0.40 | 0.50 | 0.70 | 0.10 | 12.5 | 30 | |
| 1N5822US | 0.40 | 0.50 | 0.70 | 0.10 | 12.5 | | 10 |
| 1N6864 | 0.50 | 0.70 | N/A | 0.15 | 18.0 | 30 | |
| 1N6864US | 0.50 | 0.70 | N/A | 0.15 | 18.0 | | 10 |

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

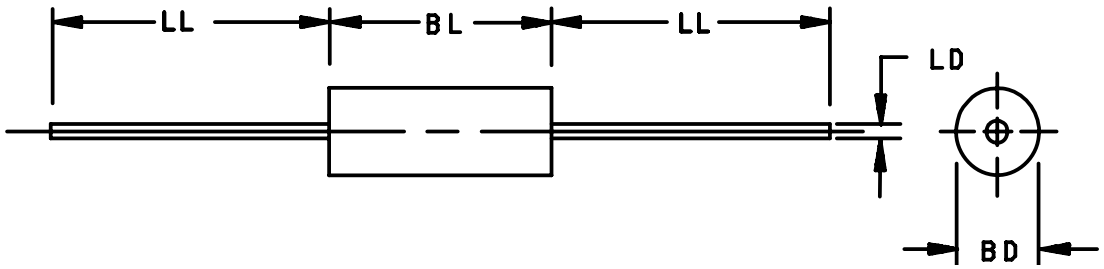
MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.



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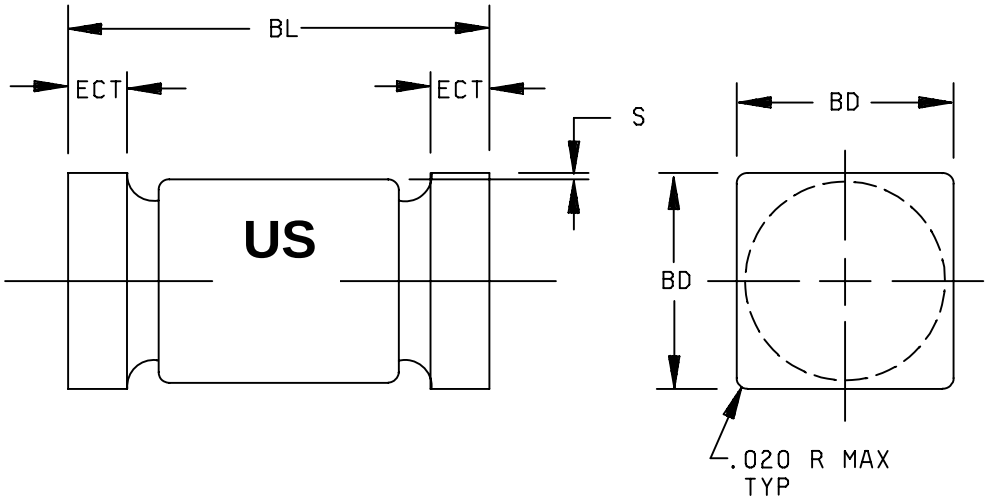

| Dimensions | | | | |
|---|---|---|---|---|
| Symbol | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| BD | .115 | .145 | 2.92 | 3.68 |
| BL | .130 | .195 | 3.30 | 4.95 |
| LD | .036 | .042 | 0.91 | 1.07 |
| LL | .900 | 1.300 | 22.86 | 33.02 |

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. In accordance with ASME Y14.5M, diameters are equivalent to Φx symbology.

FIGURE 1. Physical dimensions of 1N5822, and 1N6864.



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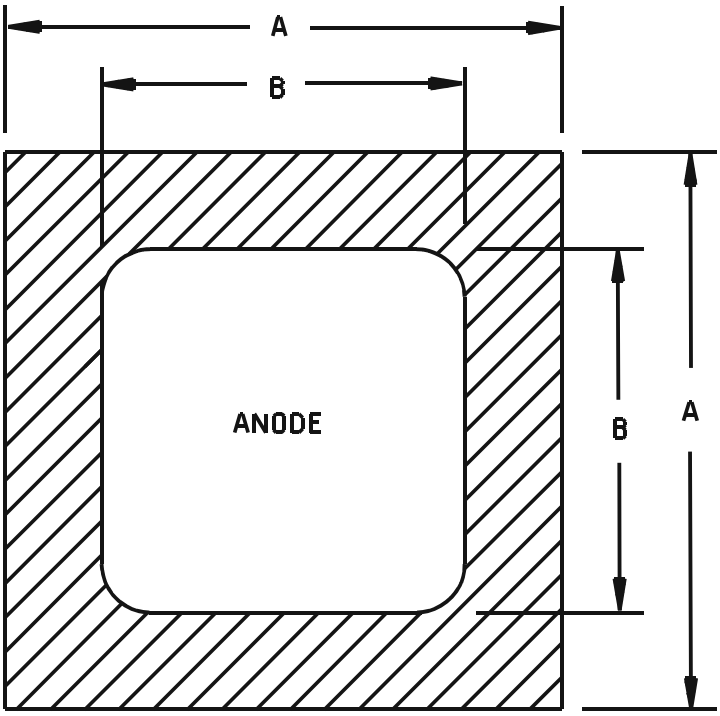

| Symbol | Dimensions | | | |
|---|---|---|---|---|
| | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| BD | .137 | .148 | 3.48 | 3.76 |
| BL | .200 | .225 | 5.08 | 5.72 |
| ECT | .019 | .028 | 0.48 | 0.71 |
| S | .003 | | 0.08 | |

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. In accordance with ASME Y14.5M, diameters are equivalent to Φx symbology.

FIGURE 2. Physical dimensions of surface mount family, 1N5822US, and 1N6864US.



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| Symbol | Dimensions | | | |
|---|---|---|---|---|
| | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| A | .062 | .064 | 1.57 | 1.63 |
| B | .052 | .054 | 1.32 | 1.37 |

Design data

Metallization:
Top: (Anode) Al
Back: (Cathode) Au

Al thickness 25,000 Å minimum.
Gold thickness 4,000 Å minimum.
Chip thickness .010 inch (0.254 mm) ± .002 (±.051).

FIGURE 3. JANC (A-version) die dimensions.



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3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figures 1 (axial leads), 2 (surface mount), and 3 (die).

3.4.1 Lead material and finish. Lead material shall be copper clad steel with a minimum of 70 percent copper by weight. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Diode construction. These devices shall be metallurgically bonded-thermally-matched-noncavity-double plug construction, utilizing a category I or III bond, in accordance with MIL-PRF-19500, except for JANHC and JANKC.

3.4.2.1 Surface mount. The surface mount ‘US’ version shall be considered structurally identical to the non-surface mount version except for lead attach.

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500. No color coding shall be permitted for part numbering.

3.5.1 Marking for surface mount (‘US’) devices. For ‘US’ version devices only, all marking, except polarity may be omitted from the body, but shall be retained on the initial container. Polarity marking of ‘US’ devices shall consist as a minimum, a band or three contrasting dots around the periphery of the cathode.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.7 Electrical test requirements. The electrical test requirements shall be as specified in table I herein.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).



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4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or requalification only. In case qualification was awarded to a prior revision of the specification sheet that did not require the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

4.2.2 JANHC and JANKC devices. Qualification for JANHC and JANKC devices shall be in accordance with MIL-PRF-19500. This testing may be performed in a TO-5 package in lieu of the axial leaded package.

* 4.3 Screening (JANTX, JANTXV, and JANS levels only). Screening shall be in accordance with table E-IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

| Screen (see table E-IV of MIL-PRF-19500) | Measurement | |
|---|---|---|
| | JANS level | JANTXV and JANTX level |
| 2 | Not required | Not required |
| 3b<br>(1) 3c | Not applicable<br>Required (see 4.3.3) | Not applicable<br>Required (see 4.3.3) |
| 4, 5, and 6 | Not applicable | Not applicable |
| 8 | Required | Not required |
| 9 | Required<br>$I_{R1}$ and $V_{FM2}$ | Not applicable |
| (2) 10 | Required<br>1N5822, $T_A$ = +90°C; $V_{RWM}$ = 40 V(pk);<br>1N6864, $T_A$ = +80°C; $V_{RWM}$ = 80 V(pk);<br>$V_{RWM}$ = half sine wave, f = 60Hz | Required<br>1N5822, $T_A$ = +90°C; $V_{RWM}$ = 40 V(pk);<br>1N6864, $T_A$ = +80°C; $V_{RWM}$ = 80 V(pk);<br>$V_{RWM}$ = half sine wave, f = 60Hz |
| 11 | Required<br>$\Delta I_{R1} \leq$ 100 percent of initial reading or 0.05 mA whichever is greater;<br>$\Delta V_{FM2} \leq \pm$50 mV dc. | Required<br>$I_{R1}$ and $V_{FM2}$ |
| 12 | See 4.3.2. | t = 48 hours. See 4.3.2. |
| 13 | Required<br>Subgroup 2 of table I herein;<br>$\Delta I_{R1} \leq$ 100 percent of initial reading or 0.05 mA whichever is greater;<br>$\Delta V_{FM2} \leq \pm$50 mV dc | Required<br>Subgroup 2 of table I herein;<br>$\Delta I_{R1} \leq$ 100 percent of initial reading or 0.05 mA whichever is greater; $V_{FM2} \leq \pm$50 mV dc |

(1) Thermal impedance shall be performed anytime after temperature cycling, screen 3a, and does not need to be repeated in screening.

(2) Junction temperature ($T_J$) is not to exceed 115°C at $V_{RWM}$. $T_J$ is affected by the device mounting thermal resistance when parasitic power is generated by the temperature dependent leakage current. Until this leakage becomes significant near thermal runaway, $T_J$ remains approximately equal to $T_A$ or $T_J$ for $I_O$ = 0.



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4.3.1 Screening (JANHC or JANKC). Screening of die shall be in accordance with MIL-PRF-19500. As a minimum, die shall be 100-percent probed in accordance with table I, subgroup 2, except for thermal impedance.

4.3.1.1 JAN testing. JAN level product will have temperature cycling and thermal impedance testing performed in accordance with MIL-PRF-19500, JANTX level screening level requirements. Electrical testing shall be in accordance with table I, subgroup 2 herein.

* 4.3.2 Burn-in conditions. Burn-in conditions are as follows: $I_F$ = 3.0 A dc (min). Mounting and test conditions shall be in accordance with method 1038 of MIL-STD-750, test condition B.

4.3.3 Thermal impedance measurements. The thermal impedance measurements shall be performed in accordance with method 3101 or 4081 of MIL-STD-750, as applicable, using the guidelines in that method for determining $I_H$ and $I_M$. The thermal impedance limit ($Z_{\Theta JX}$) shall be less than the process determined statistical maximum limit as outlined in method 3101 or 4081 of MIL-STD-750, as applicable. See group E, subgroup 4 of table II herein.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with table E-V of MIL-PRF-19500, and table I herein. The following test conditions shall be used for $Z_{\Theta JX}$, group A inspection:

a. IM measurement current 1 mA to 10 mA.

b. IH forward heating current 3A.

c. tH heating time 10 ms.

d. $t_{MD}$ measurement delay time 70 μs maximum.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in tables E-VIa and E-VIb (JANS, JANTXV, JANTX, and JAN) of MIL-PRF-19500 and as follows. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

* 4.4.2.1 Group B inspection, table E-VIa (JANS) of MIL-PRF-19500.

| Subgroup | Method | Condition |
|---|---|---|
| B3 | 1056 | -55°C to 100°C, 25 cycles, n = 22, c = 0. |
| B3 | 1051 | -55°C to 150°C, 100 cycles, n = 22, c = 0. |
| B3 | 4066 | $I_{FSM}$ = 80 A (pk), condition A 2, $I_O$ = 3 A dc; $T_A$ = room ambient as defined in 4.5 of MIL-STD-750; five surges of 8.3 ms each at 1 minute intervals. |
| B4 | 1037 | $I_F$ = 3.0 A dc; $T_A$ = room ambient as defined in the general requirements of MIL-STD-750; $t_{on}$ = $t_{off}$ = 3 minutes minimum for 2,000 cycles. |
| B5 | 1026 | $I_F$ = 3 A dc minimum, adjust $I_F$ or $T_A$ to achieve $T_J$ = +125°C minimum. |



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4.4.2.2 Group B inspection, table E-VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500.

| Subgroup | Method | Condition |
|---|---|---|
| B2 | 1056 | -55°C to 100°C, 10 cycles, n = 22, c = 0. |
| B2 | 1051 | -55°C to 150°C, 25 cycles, n = 22, c = 0. |
| B2 | 4066 | $I_{FSM}$ = 80 A (pk), condition A 2, $I_O$ = 3 A dc; $T_A$ = room ambient as defined in 4.5 of MIL-STD-750; five surges of 8.3 ms each at 1 minute intervals. |
| B3 | 1027 | $I_F$ = 3 A dc minimum, adjust $I_F$ or $T_A$ to achieve $T_J$ = +125°C. |
| B4 | 2075 | As applicable. |

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

4.4.3.1 Group C inspection, table E-VII of MIL-PRF-19500.

| Subgroup | Method | Condition |
|---|---|---|
| C2 | 2036 | Tension: Test condition A; weight = 20 pounds; t = 15 seconds.<br>Lead fatigue: Test condition E; weight 1 pound.<br>NOTE: Neither tension nor lead fatigue are applicable for US devices. |
| C5 | 4081 | See 4.4.5 herein. |
| C6 | 1027 | $I_F$ = 3 A dc minimum, adjust $I_F$ or $T_A$ to achieve $T_J$ = +125°C minimum. |

* 4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the tests and conditions specified for subgroup testing in table E-IX of MIL-PRF-19500, and table II herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

* 4.4.5 Thermal resistance. Thermal resistance measurement shall be in accordance with method 3101 or 4081 of MIL-STD-750. Forced moving air or draft shall not be permitted across the device during test. The maximum limit for $R_{\Theta JL}$ under these test conditions shall be $R_{\Theta JL}$ (max) = 30°C/W, $R_{\Theta JEC}$ (max) = 10°C/W. The following conditions shall apply when using method 3101:

a. $I_M$ 1mA to 10mA.

b. $I_H$ 3A minimum.

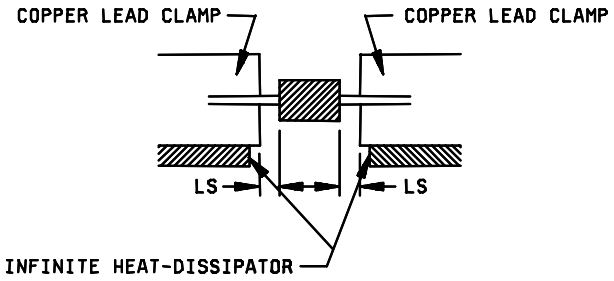
c. $t_H$ 25 seconds minimum.

d. $t_{MD}$ 70 μs maximum.



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LS = lead spacing = .375 inch (9.53 mm) for non-surface mount devices and 0 inch (0 mm) for surface mount devices as defined as follows:




4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Steady-state operation life. This test shall be conducted with a half-sine wave of the specified peak voltage impressed across the diode in the reverse direction followed by a half-sine waveform of the specified average rectified current. The forward conduction angle of the rectified current shall not be greater than 180 degrees nor less than 150 degrees.



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TABLE I. Group A inspection.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 1 | | | | | | |
| Visual and mechanical examination | 2071 | | | | | |
| Subgroup 2 | | | | | | |
| Thermal impedance | 3101 | See 4.3.3 | $Z_{\Theta JX}$ | | 2.5 | °C/W |
| Forward voltage | 4011 | $I_{FM}$ = 1.0 A (pk) pulse method (see 4.5.1) | $V_{FM1}$ | | | |
| 1N5822, 1N5822US | | | | | 0.40 | V |
| 1N6864, 1N6864US | | | | | 0.50 | V |
| 1N5822, 1N5822US | 4011 | $I_{FM}$ = 3.0 A (pk) pulse method (see 4.5.1) | $V_{FM2}$ | | 0.50 | V |
| 1N6864, 1N6864US | | | | | 0.70 | V |
| 1N5822, 1N5822US | 4011 | $I_{FM}$ = 9.4 A (pk) pulse method (see 4.5.1) | $V_{FM3}$ | | 0.70 | V |
| Reverse current leakage | 4016 | | $I_{RM1}$ | | | |
| 1N5822, 1N5822US | | $V_{RM}$ = 40 V (pk) pulse method | | | 0.10 | mA |
| 1N6864, 1N6864US | | $V_{RM}$ = 80 V (pk) pulse method (see 4.5.1) | | | 0.15 | mA |
| Subgroup 3 | | | | | | |
| High temperature operation: | | $T_A$ = +100°C | | | | |
| Reverse current leakage | 4016 | | $I_{RM2}$ | | | |
| 1N5822, 1N5822US | | $V_{RM}$ = 40 V (pk) pulse method | | | 12.5 | mA |
| 1N6864, 1N6864US | | $V_{RM}$ = 80 V (pk) pulse method (see 4.5.1) | | | 18.0 | mA |
| Forward voltage | 4011 | $I_F$ = 3.0 A (pk) pulse method (see 4.5.1) | $V_{FM4}$ | | | |
| 1N5822, 1N5822US | | | | | 0.47 | V |
| 1N6864, 1N6864US | | | | | 0.65 | V |
| Low temperature operation: | | $T_A$ = -55°C | | | | |
| Reverse current leakage | 4016 | | $I_{RM3}$ | | | |
| 1N5822, 1N5822US | | $V_{RM}$ = 40 V (pk) pulse method | | | 0.40 | mA |
| 1N6864, 1N6864US | | $V_{RM}$ = 80 V (pk) pulse method (see 4.5.1) | | | 0.55 | mA |
| Forward voltage | 4011 | $I_F$ = 3.0 A (pk) pulse method (see 4.5.1) | $V_{FM5}$ | | | |
| 1N5822, 1N5822US | | | | | 0.62 | V |
| 1N6864, 1N6864US | | | | | 0.80 | V |
| Subgroup 4, 5, 6, and 7 | | | | | | |
| Not applicable | | | | | | |

1/ For sampling plan, see MIL-PRF-19500.



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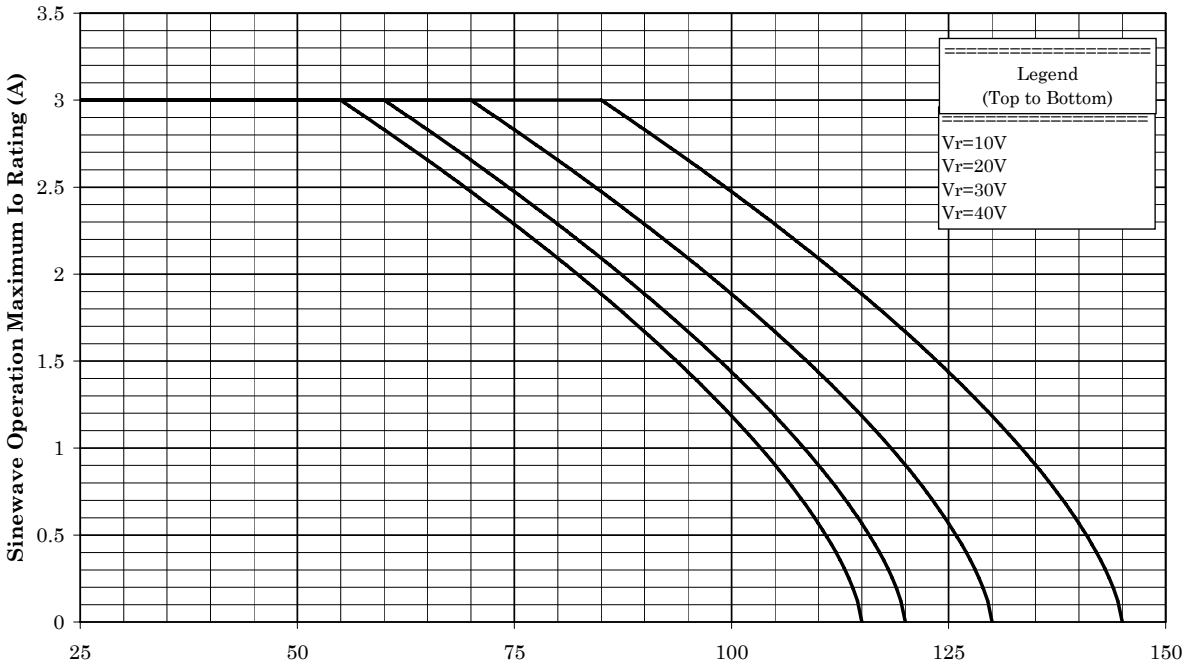
TABLE II. Group E inspection (all quality levels) for qualification and requalification.

| Inspection | MIL-STD-750 | | Sampling plan |
|---|---|---|---|
| | Method | Conditions | |
| Subgroup 1 | | | n = 22, c = 0 |
| Temperature cycling | 1051 | -65°C to 150°C, 500 cycles | |
| Hermetic seal | 1071 | Test condition E | |
| Electrical measurement | | See table I, subgroup 2 | |
| Subgroup 2 | | | |
| Intermittent Operating Life | 1036 | 10,000 cycles | n = 22, c = 0 |
| Electrical measurement | | See table I, subgroup 2 | |
| Subgroup 4 | | | |
| Thermal impedance curves | | See MIL-PRF-19500 | |
| Subgroup 5 | | | |
| Not applicable | | | |
| Subgroup 6 | | | n = 3 |
| ESD | 1020 | | |
| Subgroup 8 | | | n = 45 |
| Resistance to glass cracking | 1057 | Test to destruction or 25 cycles max, whichever comes first. | |



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**TEMPERATURE-CURRENT DERATING CURVE**
**1N5822 AXIAL LEAD-MOUNTED**




Sinewave Operation 50% Duty Cycle $T_L$ (°C) (Lead = 3/8")
$R_{\Theta JL}$ = 30.0°C/W

NOTES:

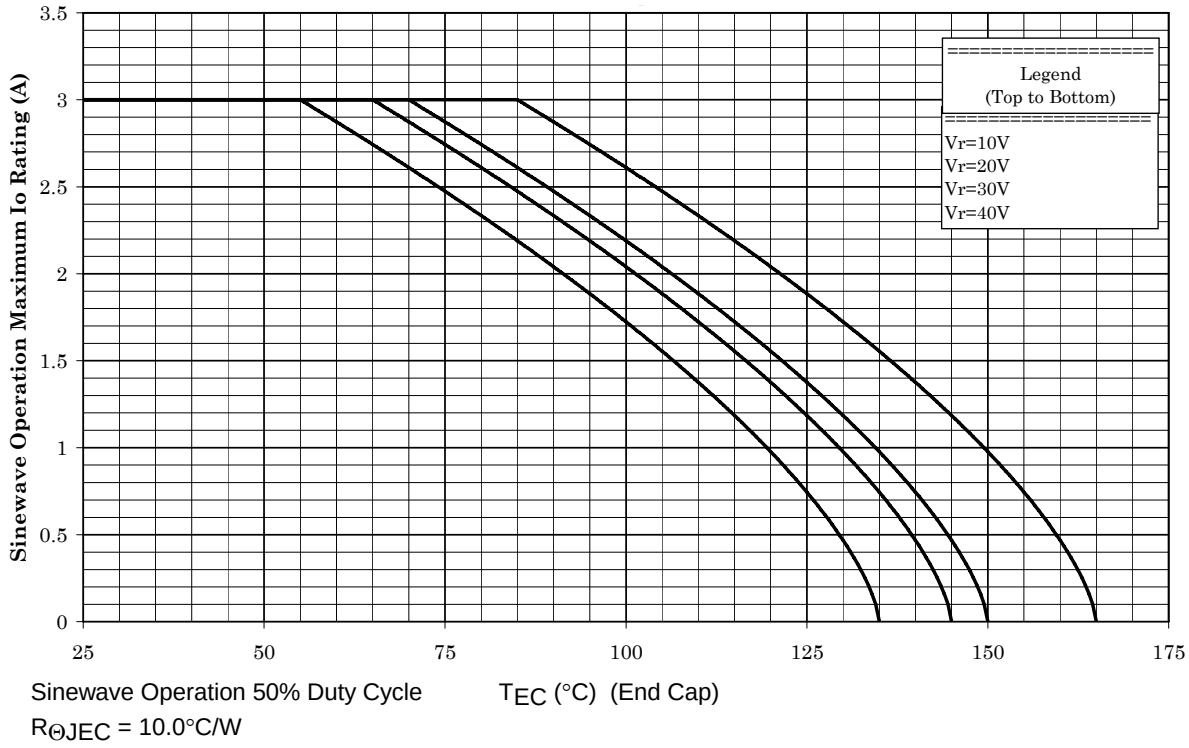
1. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating ≤ $T_J$ specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum $T_J$ allowed.
2. This temperature-current derating curve varies with applied voltage.

FIGURE 4. Temperature current derating for 1N5822.



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**TEMPERATURE-CURRENT DERATING CURVE**
**1N5822US MELF END-CAP MOUNTED**




NOTES:

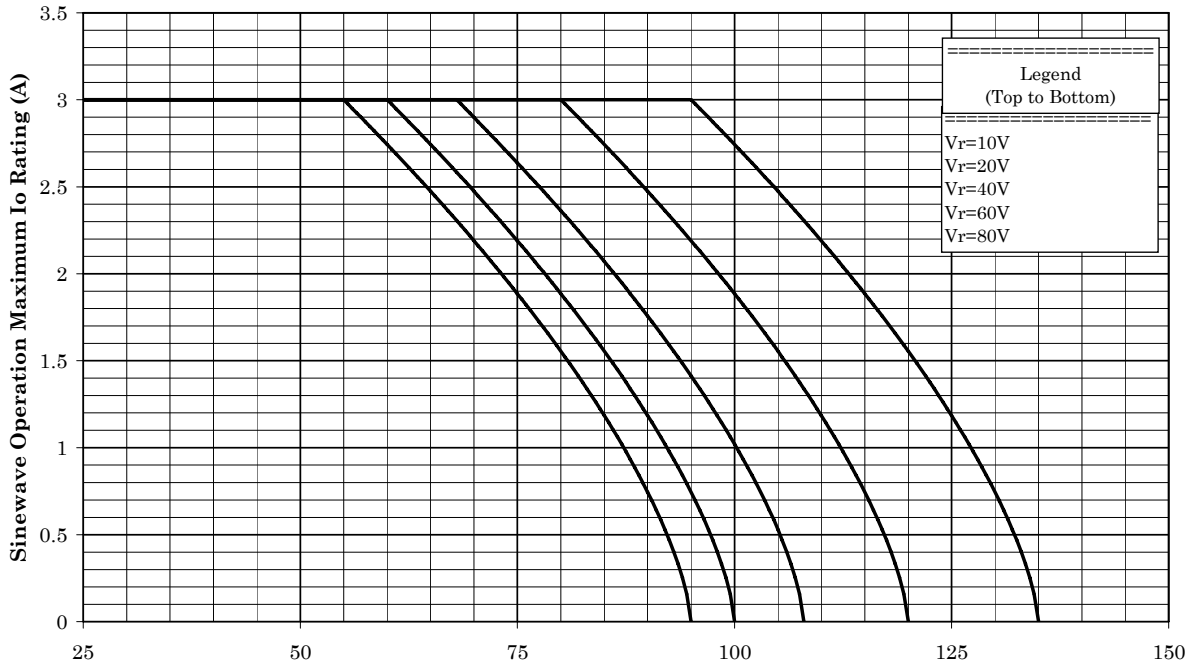
1. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating ≤ $T_J$ specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum $T_J$ allowed.
2. This temperature-current derating curve varies with applied voltage.

FIGURE 5. Temperature current derating for 1N5822US.



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**TEMPERATURE-CURRENT DERATING CURVE**
**1N6864 AXIAL LEAD-MOUNTED**




Sinewave Operation 50% Duty Cycle $T_L$ (°C) (Lead = 3/8")
$R_{\Theta JL}$ = 30.0°C/W

NOTES:

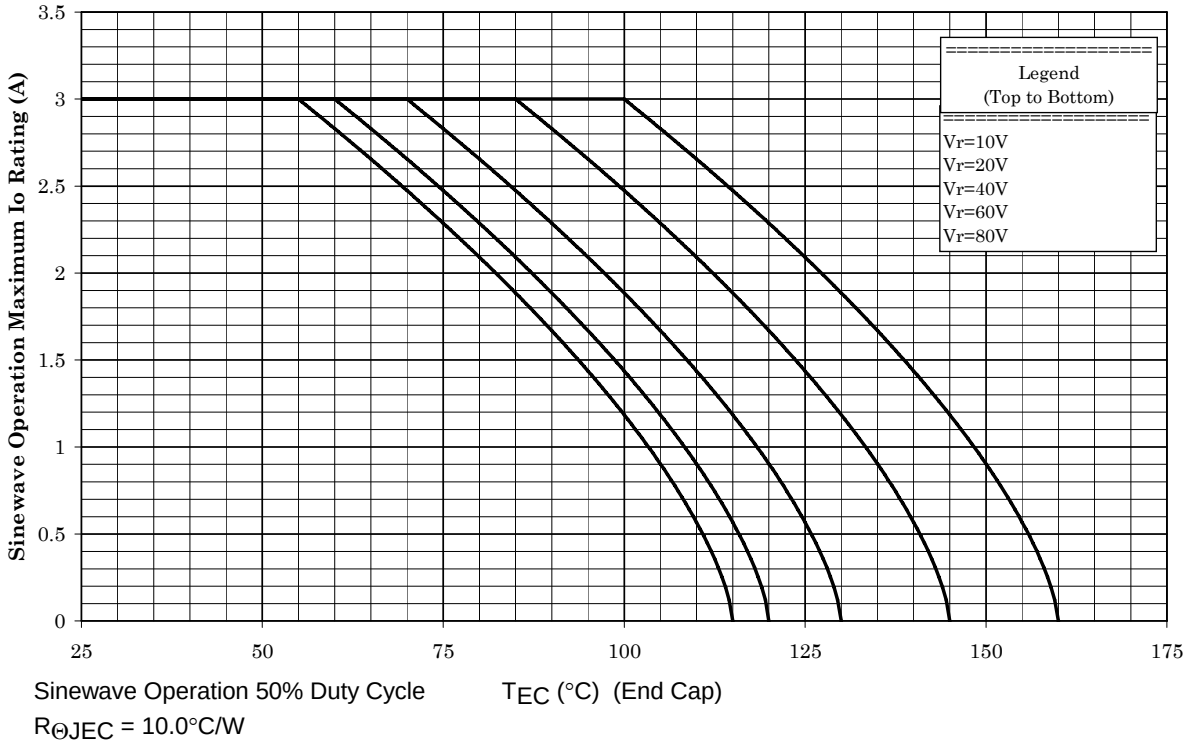
1. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating ≤ $T_J$ specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum $T_J$ allowed.
2. This temperature-current derating curve varies with applied voltage.

FIGURE 6. Temperature current derating for 1N6864.



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**TEMPERATURE-CURRENT DERATING CURVE**
**1N6864US MELF END-CAP MOUNTED**




NOTES:

1. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating ≤ $T_J$ specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum $T_J$ allowed.
2. This temperature-current derating curve varies with applied voltage.

FIGURE 7. Temperature current derating for 1N6864US.



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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

* 6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. The acquisition requirements should specify the following.

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. Product assurance level and type designator.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML-19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Columbus, ATTN: DSCC-VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil .

6.4 Suppliers of die. The qualified die suppliers with the applicable letter version (e.g., JANHCA1N5822) will be identified on the QML.

| JANC ordering information | |
|---|---|
| PIN | Manufacturer |
| | 43611 |
| 1N5822 | JANHCA1N5822<br>JANKCA1N5822 |
| 1N6864 | JANHCA1N6864<br>JANKCA1N6864 |



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6.5 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - EC
Air Force - 11
NASA - NA
DLA - CC

Review activities:
Army - AR, MI, SM
Navy - AS, MC
Air Force - 19

Preparing activity:
DLA - CC

(Project 5961-2007-010)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil .



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